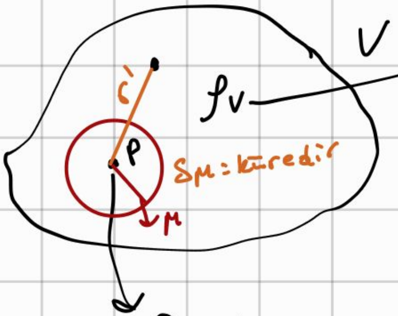
# Yüzeysel yüke Etki eden kuvvet

$q_1$ ----- $q_n$ tane yük olsun.
$q_1$ yüküne etki eden kuvveti

$$\vec{F}_1 = \frac{q_1}{4\pi\varepsilon} \sum_{i=2}^{n} q_i \frac{\vec{r_i}}{r_i^3}$$

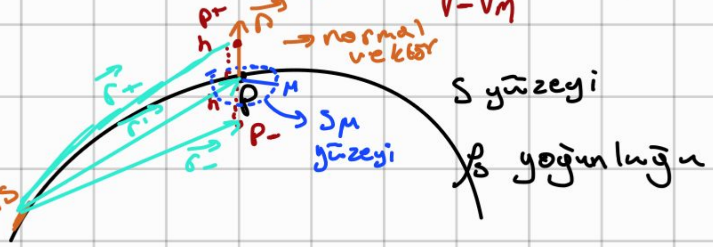


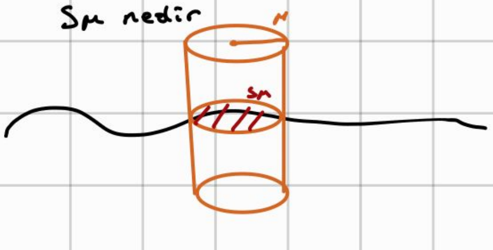
yük yoğunluğu ile yükleri dağıtalım

P noktasına etki eden kuvveti hesaplayalım

Sμ küresi dışında bulunan bütün yükleri içerideki birim yüke etki kuvvet (Elektrik alan) Bu kuvveti hesaplayıp, $\mu \to 0$ götürülürse P noktasındaki elektrik alanı buluruz.

$$\vec{E}(P) = \frac{1}{4\pi\varepsilon} \lim_{\mu \to 0} \int_{V-V_\mu} \rho_v \frac{\vec{r'}}{r'^3}\, dV$$





P noktası üzerindeki elektrik alanı bulalım.

$$\vec{E}(P) = \frac{1}{4\pi\varepsilon} \lim_{\mu \to 0} \int_{S-S_\mu} \rho_s \frac{\vec{r'}}{r'^3}\, ds$$

Daha basit bir yol bulmaya çalışalım

$$\vec{E}(P_+) = \frac{1}{4\pi\varepsilon} \int_S \rho_s \frac{\vec{r_+}}{r_+^3}\, ds$$

$$\vec{E}(P_-) = \frac{1}{4\pi\varepsilon} \int_S \rho_s \frac{\vec{r_-}}{r_-^3}\, ds$$

$$\vec{E}(P_+) = \frac{1}{4\pi\varepsilon}\int_{S-S_\mu} \rho_s \frac{\vec{r}_+}{r_+^3}\, ds + \overbrace{\frac{1}{4\pi\varepsilon}\int_{S_\mu} \rho_s \frac{\vec{r}_+}{r_+^3}\, ds}^{e(h,\mu)\hat{n}}$$

$$\vec{E}(P_-) = \frac{1}{4\pi\varepsilon}\int_{S-S_\mu} \rho_s \frac{\vec{r}_-}{r_-^3}\, ds + \underbrace{\frac{1}{4\pi\varepsilon}\int_{S_\mu} \rho_s \frac{\vec{r}_-}{r_-^3}\, ds}_{-e(h,\mu)\hat{n}}$$

(yani h'e 0 ra götürürsek)

$$\lim_{P_+\to P} \vec{E}(P_+) = \vec{E}(P) + e(0,\mu)\hat{n}$$

$$\lim_{P_-\to P} \vec{E}(P_-) = \vec{E}(P) - e(0,\mu)\hat{n}$$

$$\vec{E}^+(P) = \vec{E}(P) + e(0,\mu)\hat{n}$$
$$\vec{E}^-(P) = \vec{E}(P) - e(0,\mu)\hat{n}$$

$\mu \to 0$ ra götürürsek

$$\vec{E}^+(P) = \vec{E}(P) + e(0)\hat{n}$$
$$\vec{E}^-(P) = \vec{E}(P) - e(0)\hat{n}$$
$$+\overline{\qquad\qquad\qquad\qquad}$$
$$\vec{E}(P) = \frac{1}{2}\left[\vec{E}^+(P) + \vec{E}^-(P)\right]$$

Herhangi bir $\rho_s ds$ yüzeyel yüküne etki eden kuvvet:

$$d\vec{F} = \frac{1}{2}\rho_s ds\left[\vec{E}^+ + \vec{E}^-\right]$$

Sayfa 55 problem 1

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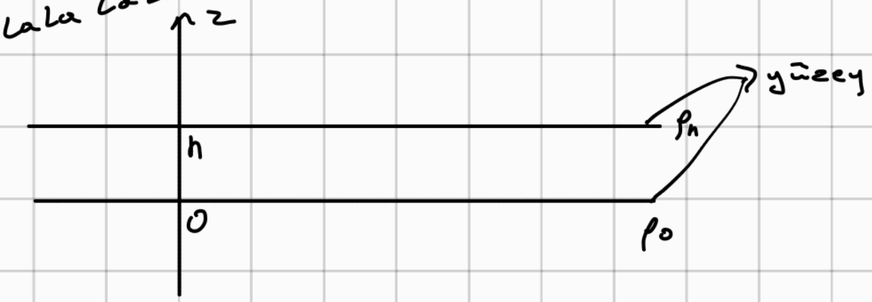

a) z=0'daki ve z=h'taki elektrik alanı doğrudan bulunuz.

sadece $\rho_H$'ın etkisi var diyelim

$$\vec{E}(0) = -\frac{\rho_h}{2\varepsilon}\vec{e}_z \qquad \vec{E}(h) = +\frac{\rho_0}{2\varepsilon}\vec{e}_z$$

b) $z>h$, $0<z<h$, $z<0$

$$\vec{E} = \frac{(\rho_0+\rho_h)}{2\varepsilon}\vec{e}_z$$

$0 < z < h$

$$\vec{E} = \frac{(\rho_0 - \rho_n)}{2\varepsilon}\vec{e}_z$$

$z < 0$

$$\vec{E} = -\frac{(\rho_0 + \rho_n)}{2\varepsilon}\vec{e}_z$$

c) $z=0$ ve $z=h$

$$\vec{E}(0) = \frac{1}{2}\left[\vec{E}^{+}(0) + \vec{E}^{-}(0)\right]$$

$$\vec{E}^{+}(0) = \frac{(\rho_0 - \rho_n)}{2\varepsilon}\vec{e}_z$$

$$\vec{E}^{-}(0) = -\frac{(\rho_0 + \rho_n)}{2\varepsilon}\vec{e}_z$$

$$\vec{E}(0) = \frac{1}{2}\left[\frac{\rho_0 - \rho_n - \rho_0 - \rho_n}{2\varepsilon}\vec{e}_z\right]$$

$$= -\frac{\rho_n}{2\varepsilon}\vec{e}_z$$

$$\vec{E}(h) = \frac{1}{2}\left[\vec{E}^{+}(h) + \vec{E}^{-}(h)\right]$$

$$\vec{E}^{+}(h) = \frac{\rho_0 + \rho_n}{2\varepsilon}\vec{e}_z$$

$$\vec{E}^{-}(h) = \frac{(\rho_0 - \rho_n)}{2\varepsilon}\vec{e}_z$$

$$\vec{E}(h) = \frac{1}{2}\left[\frac{\rho_0 + \rho_n + \rho_0 - \rho_n}{2\varepsilon}\right]\vec{e}_z$$

$$\vec{E}(h) = \frac{\rho_0}{2\varepsilon}\vec{e}_z$$

d) $z=0$ üzerinde $S$ alan parçasına etki eden kuvvet

$$\vec{F} = \frac{1}{2}\rho_0 S\left[\vec{E}^{+}(0) + \vec{E}^{-}(0)\right]$$

$$= \frac{-\rho_0 \rho_n S}{2\varepsilon}\vec{e}_z$$

e) $\rho_n = \rho_0 \Rightarrow \vec{F} = \frac{-\rho_0^2}{2\varepsilon}\vec{e}_z$

$\rho_n = -\rho_0 \Rightarrow \vec{F} = \frac{\rho_0^2}{2\varepsilon}\vec{e}_z$